Supplementary Material

History-Aware Control Barrier Functions with Nonsingular Fading-Memory Kernels

Ege C. Altunkaya, Esra Demir, and İbrahim Özkol

1 Purpose

This supplementary document collects numerical evidence that complements the six-page manuscript. The manuscript remains self-contained: no theorem, assumption, or safety guarantee depends on this document. The supplementary study has two purposes. First, it tests whether the reported behavior persists over a structured validation grid rather than only the representative twin-square trajectory shown in the paper. Second, it evaluates the nonsingular-kernel construction beyond the Caputo–Fabrizio (CF) exponential specialization using a bi-exponential kernel with an exact two-state realization and a Gaussian kernel evaluated directly from retained barrier history.

All controller parameters within a given study are fixed across the corresponding validation cases. No cross-kernel performance-optimality claim is made; the non-CF studies are intended to test the realization-independent history-aware mechanism.

2 Extended Stress-Test Validation

2.1 High-demand sweep

The high-demand sweep contains all $4 \times 3 \times 3 = 36$ combinations of square command

$$n_{z,\text{sq}} \in \{2.4, 2.6, 2.8, 3.0\}\ \text{g}, \qquad T_r \in \{2, 3, 4\}\ \text{s},$$

and prepared history $H \in \{\text{adverse}, \text{neutral}, \text{favorable}\}$. CF-FOCBF preserves the hard incidence constraint, the prescribed history-conditioned policy, and QP feasibility in all 36 cases. The performance-optimal state-only CBF remains hard-safe but violates the prescribed policy in all 36 cases, while the unfiltered nominal response violates the hard incidence constraint in all 36 cases.

Relative to the policy-preserving state-only CBF, CF-FOCBF reduces tracking RMSE in every high-demand case. Across the 36 cases, the mean reduction is 30.938%, the median is 31.755%, and the observed range is 23.360–35.773%.

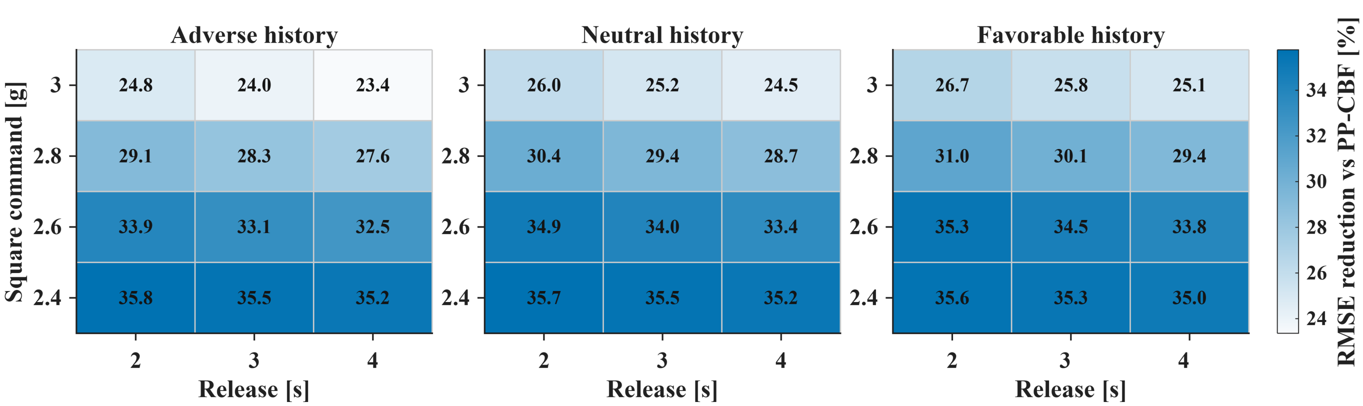

Figure 1: Complete high-demand RMSE-reduction atlas. Each cell reports the percentage reduction of CF-FOCBF relative to the history-blind policy-preserving CBF for the same command level, release duration, and prepared history. Improvement is observed in all 36 tested cases.

2.2 Safety audit and low-demand negative controls

A separate $3 \times 2 \times 3 = 18$-case low-demand sweep uses square commands $\{1.0, 1.4, 1.8\}$ g and release durations $\{2, 4\}$ s. CF-FOCBF remains inactive in all 18 cases and therefore coincides with the nominal controller. The policy-preserving state-only CBF intervenes in all 18 cases, with a maximum input modification of approximately 13.13°.

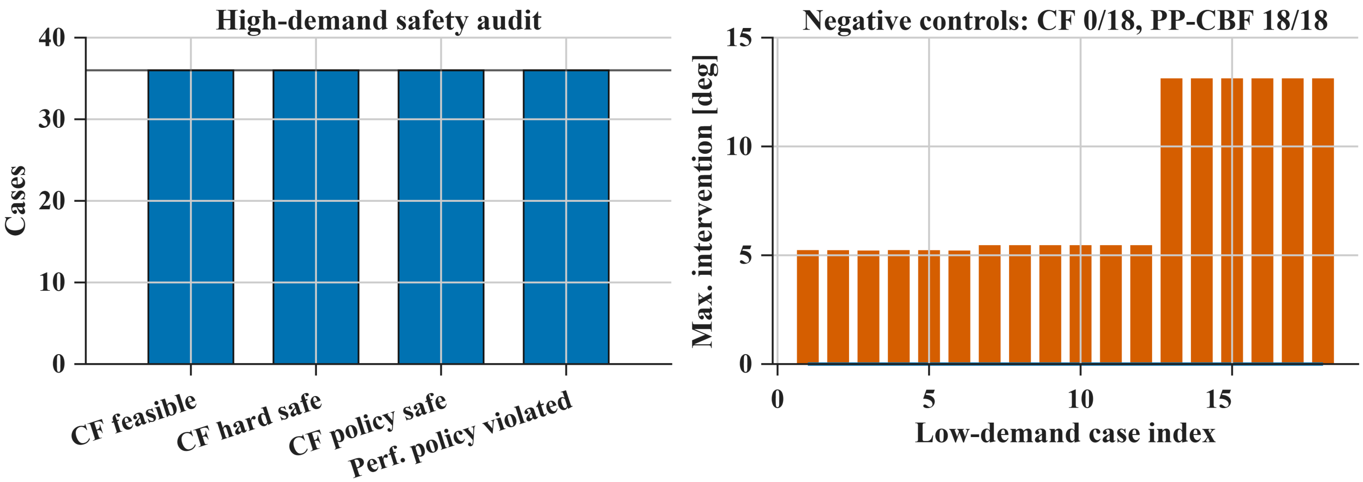

Figure 2: Left: high-demand safety and feasibility audit. Right: low-demand negative controls. CF-FOCBF is inactive in all 18 low-demand cases, whereas the policy-preserving state-only CBF intervenes in every case.

3 Beyond CF: Nonsingular-Kernel Validation

The generalized construction is stated in the convolution domain and does not require a finite-dimensional realization. To separate that result from the CF implementation, the same plant, nominal controller, preparation protocol, activation state, actuator limits, reserve policy, and twin-square demand are repeated with two additional kernels.

3.1 Realization hierarchy

The three kernels used in the validation form the hierarchy

$$\kappa_{\mathrm{CF}}(t) = K_0 e^{-\lambda t}, \tag{1}$$

$$\kappa_{\mathrm{B}}(t) = K_0 \left[a e^{-\lambda_1 t} + (1-a) e^{-\lambda_2 t} \right], \tag{2}$$

$$\kappa_{\mathrm{G}}(t) = K_0 e^{-(t/\tau_G)^2}. \tag{3}$$

The CF kernel admits an exact one-state realization. The bi-exponential kernel admits an exact two-state realization, while the Gaussian convolution is evaluated directly from retained history rather than replaced by a finite-dimensional surrogate. The comparison therefore tests the same history-aware mechanism across three different memory realizations.

Table 1: Kernel implementations used in the extended validation.

Kernel	$\kappa(0)$	Reserve fraction θ	Implementation
CF	2	0.40	exact one-state realization
Bi-exponential	2	0.40	exact two-state realization
Gaussian	2	0.40	direct retained-history evaluation

For the bi-exponential study, $a = 5/11$, $\lambda_1 = 0.25\ \mathrm{s}^{-1}$, and $\lambda_2 = 0.8\ \mathrm{s}^{-1}$, giving the same integrated memory time as the CF reference. The Gaussian reference uses $\tau_G = 1.6$ s. These parameters are not selected to outperform CF on the twin-square maneuver; the purpose is to test whether history-conditioned authority persists beyond the one-state exponential specialization.

3.2 Bi-exponential kernel

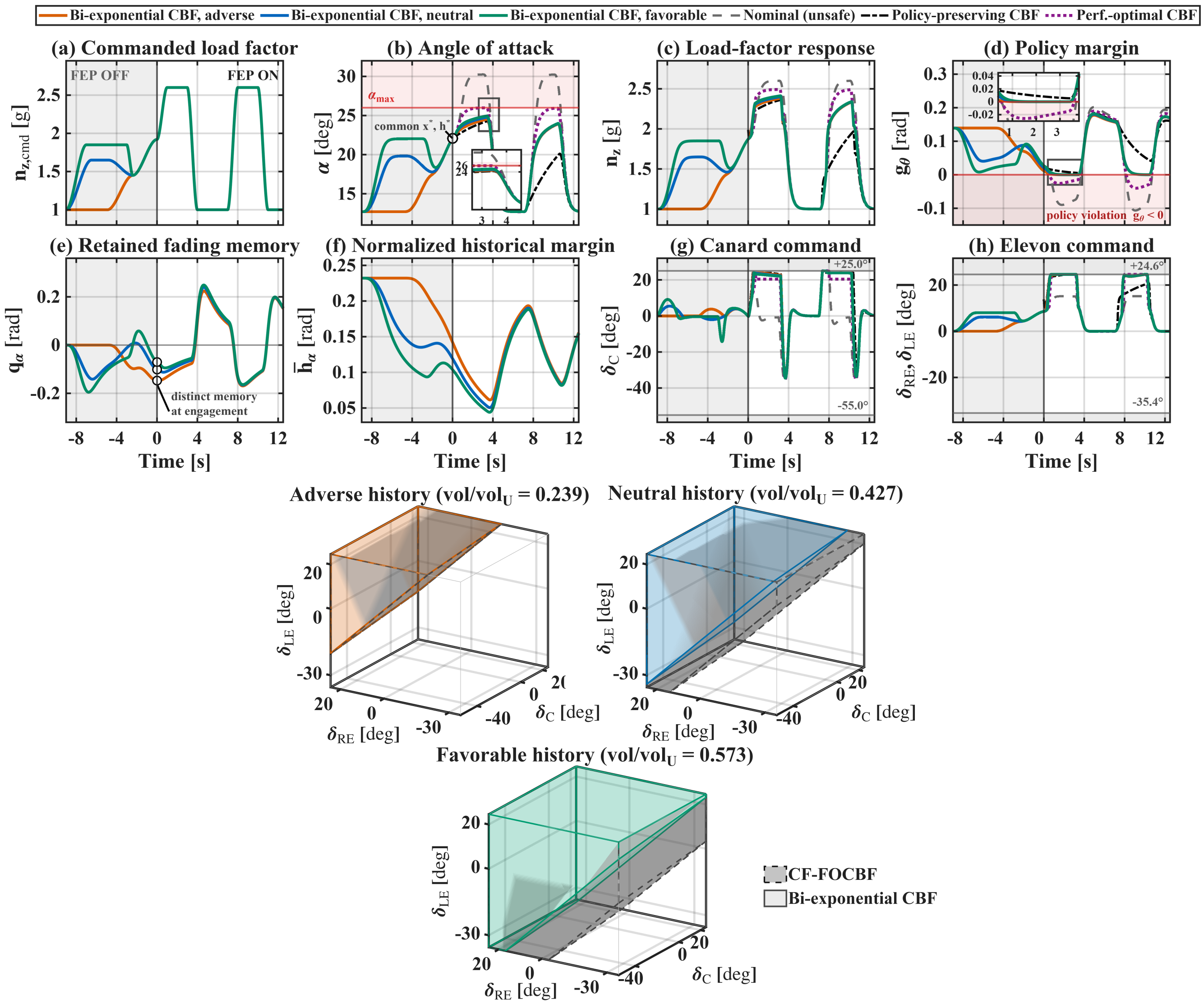

Figure 3: Bi-exponential validation under the common twin-square benchmark. Top: state, policy-margin, memory, and control responses for the three prepared histories. Bottom: same-state admissible authority, with normalized volumes 0.239, 0.427, and 0.573 for adverse, neutral, and favorable histories; the dashed geometry is the history-matched CF reference.

3.3 Gaussian kernel

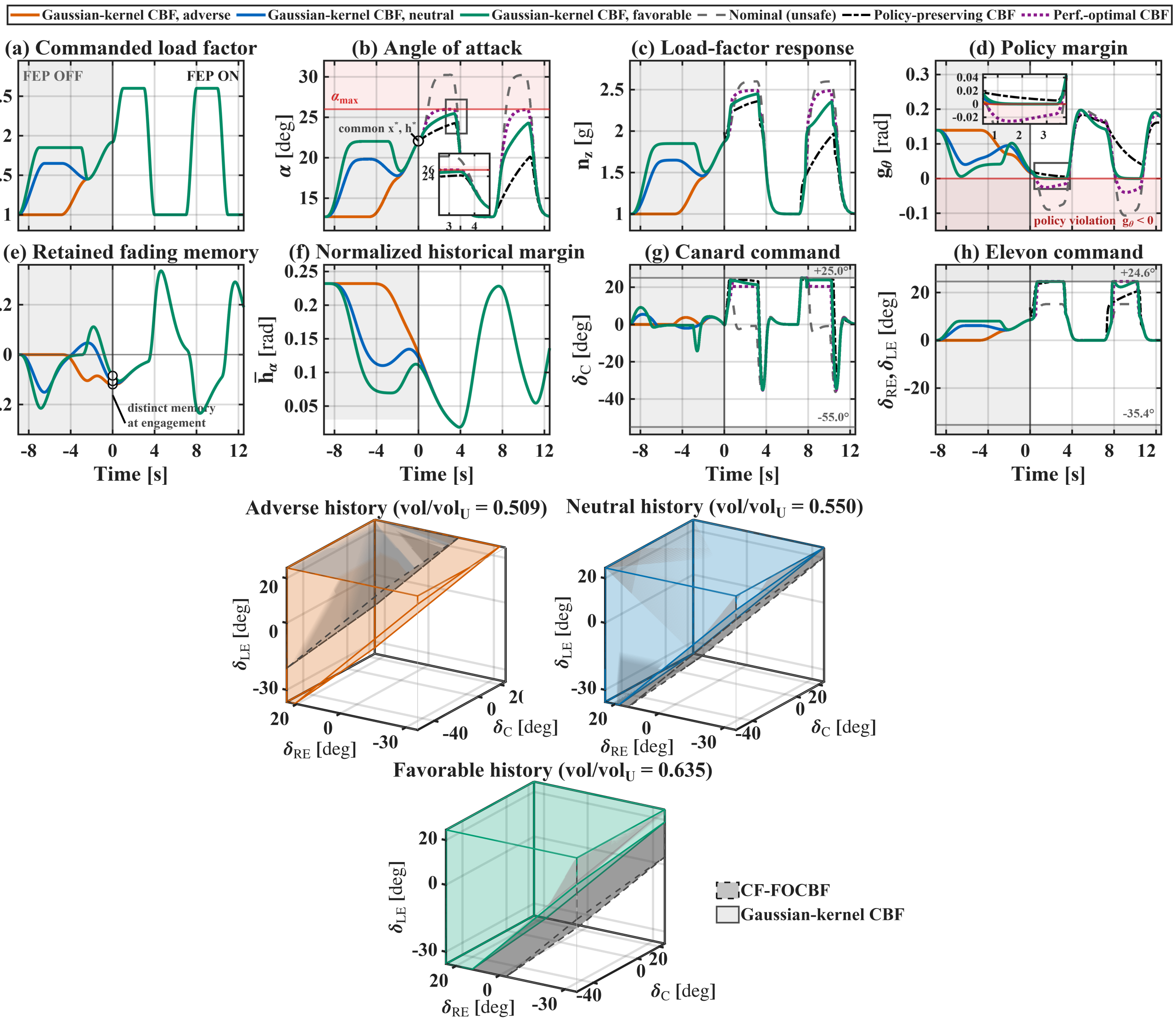

Figure 4: Gaussian-kernel validation with direct retained-history evaluation. Top: state, policy-margin, memory, and control responses under the common twin-square benchmark. Bottom: same-state admissible authority, with normalized volumes 0.509, 0.550, and 0.635 for adverse, neutral, and favorable histories; the dashed geometry is the history-matched CF reference.

3.4 Same-state authority ordering

Table 2 summarizes the normalized admissible-control volumes at the common activation state. The numerical values should not be read as a cross-kernel ranking because the kernels are not performance-matched. The relevant observation is the strict adverse < neutral < favorable ordering within each kernel.

Table 2: Normalized admissible-control volume at the common activation state.

Kernel	Adverse	Neutral	Favorable
CF	0.242	0.616	0.823
Bi-exponential	0.239	0.427	0.573
Gaussian	0.509	0.550	0.635

The strict within-kernel ordering demonstrates that the same-state admissible authority remains conditioned by retained barrier history for an exact two-state memory and for a directly evaluated non-exponential memory. This is the intended numerical validation of the generalized nonsingular-kernel formulation.

4 Availability

A cleaned and documented MATLAB reproduction package is being prepared for public release. The numerical datasets underlying the reported figures and tables are available from the corresponding author upon reasonable request. The project webpage provides the final public figures and this supplementary document.